

7.4.18

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Question

If the lines $3x - 4y - 7 = 0$ and $2x - 3y - 5 = 0$ are two diameters of a circle of area 49π square units, find the equation of the circle.

- ① $x^2 + y^2 + 2x - 2y - 47 = 0$
- ② $x^2 + y^2 + 2x - 2y - 62 = 0$
- ③ $x^2 + y^2 - 2x + 2y - 62 = 0$
- ④ $x^2 + y^2 - 2x + 2y - 47 = 0$

Solution

Given two lines are,

$$3x - 4y - 7 = 0 \quad (1)$$

$$2x - 3y - 5 = 0 \quad (2)$$

Since the lines are diameters, the center lies on both:

$$\mathbf{n}_1^T \mathbf{c} = d_1, \quad (3)$$

$$\mathbf{n}_2^T \mathbf{c} = d_2, \quad (4)$$

where

$$\mathbf{n}_1 = \begin{bmatrix} 3 \\ -4 \end{bmatrix}, \quad d_1 = 7, \quad \mathbf{n}_2 = \begin{bmatrix} 2 \\ -3 \end{bmatrix}, \quad d_2 = 5 \quad (5)$$

Solution

In matrix form,

$$\begin{bmatrix} 3 & -4 \\ 2 & -3 \end{bmatrix} \mathbf{c} = \begin{bmatrix} 7 \\ 5 \end{bmatrix} \quad (6)$$

The augmented matrix is

$$\left[\begin{array}{cc|c} 3 & -4 & 7 \\ 2 & -3 & 5 \end{array} \right] \quad (7)$$

Perform row operations to reduce it to RREF:

$$\left[\begin{array}{cc|c} 3 & -4 & 7 \\ 2 & -3 & 5 \end{array} \right] \xrightarrow{R_1 \rightarrow \frac{1}{3}R_1} \left[\begin{array}{cc|c} 1 & -\frac{4}{3} & \frac{7}{3} \\ 2 & -3 & 5 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & -\frac{4}{3} & \frac{7}{3} \\ 0 & -\frac{1}{3} & \frac{1}{3} \end{array} \right] \quad (8)$$

$$\xrightarrow{R_2 \rightarrow -3R_2} \left[\begin{array}{cc|c} 1 & -\frac{4}{3} & \frac{7}{3} \\ 0 & 1 & -1 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 + \frac{4}{3}R_2} \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -1 \end{array} \right] \quad (9)$$

Solution

Hence, the centre is

$$\mathbf{c} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad (10)$$

The general conic form of a circle is

$$g(\mathbf{x}) = \mathbf{x}^\top V \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0, \quad (11)$$

where $V = I$ for a circle. The centre satisfies

$$V\mathbf{c} + \mathbf{u} = 0 \implies \mathbf{u} = -\mathbf{c} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad (12)$$

Solution

Given the area 49π , the radius squared is

$$r^2 = 49. \quad (13)$$

Using the vector form relation for a circle:

$$r^2 = \mathbf{u}^\top \mathbf{u} - f = 2 - f \implies f = -47. \quad (14)$$

Hence, the equation of the circle in vector form is

$$\mathbf{x}^\top V \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0, \quad (15)$$

which in coordinates becomes

$$x^2 + y^2 + 2(-1)x + 2(1)y - 47 = 0, \quad (16)$$

$$\boxed{x^2 + y^2 - 2x + 2y - 47 = 0}. \quad (17)$$

Part 1

```
#include <stdio.h>
#include <math.h>

// Function to compute the circle equation parameters
void solve_circle(double *params) {
    double a1 = params[0];
    double b1 = params[1];
    double c1 = params[2];
    double a2 = params[3];
    double b2 = params[4];
    double c2 = params[5];
    double area = params[6];

    // Use direct numeric value of pi
    double pi = 3.141592653589793;
```

Part 2

```
// Compute determinant to check if lines intersect
double det = a1 * b2 - a2 * b1;
if (fabs(det) < 1e-10) {
    printf("Lines are parallel or coincident. Cannot
           determine center.\n");
    return;
}
double rhs1 = -c1;
double rhs2 = -c2;

double h = (rhs1 * b2 - b1 * rhs2) / det;
double k = (a1 * rhs2 - rhs1 * a2) / det;

if (area <= 0) {
    printf("Area must be positive.\n");
    return;
}
```


Part 3

```
double r = sqrt(area / pi);

// Compute circle coefficients
double D = -2 * h;
double E = -2 * k;
double F = h * h + k * k - r * r;

// Store results in params array
params[7] = h;
params[8] = k;
params[9] = r;
params[10] = D;
params[11] = E;
params[12] = F;
}
```

Part 4

```
int main() {  
    double params[13];  
  
    printf("Enter coefficients of first line (a1 b1 c1) for a1*x  
        + b1*y + c1 = 0:\n");  
    scanf("%lf %lf %lf", &params[0], &params[1], &params[2]);  
  
    printf("Enter coefficients of second line (a2 b2 c2) for a2*x  
        + b2*y + c2 = 0:\n");  
    scanf("%lf %lf %lf", &params[3], &params[4], &params[5]);  
  
    printf("Enter area of the circle (numeric):\n");  
    scanf("%lf", &params[6]);
```

Part 5

```
solve_circle(params);

printf("\n--- Results ---\n");
printf("Center (h, k): (%.3f, %.3f)\n", params[7], params[8])
    ;
printf("Radius: %.3f\n", params[9]);
printf("Equation: x^2 + y^2 %+.3fx %+.3fy %+.3f = 0\n",
    params[10], params[11], params[12]);

return 0;
}
```

Python Code: Import and Define points

Part 1

```
import ctypes
import os
import math
import numpy as np
import matplotlib.pyplot as plt

# Load the compiled C library
if os.name == "nt": # Windows
    lib = ctypes.CDLL("./code13.dll")
else:
    lib = ctypes.CDLL("./code13.so")

# Define argument and return types for C function
lib.solve_circle.argtypes = [ctypes.POINTER(ctypes.c_double)]
lib.solve_circle.restype = None
```

Part 2

```
# Input values (example from question)
# Lines:  $3x - 4y - 7 = 0$  and  $2x - 3y - 5 = 0$ 
# Area = 49
a1, b1, c1 = 3, -4, -7
a2, b2, c2 = 2, -3, -5
area = 49 * math.pi

# Prepare array of doubles (13 elements)
params = (ctypes.c_double * 13)()
params[0] = a1
params[1] = b1
params[2] = c1
params[3] = a2
params[4] = b2
params[5] = c2
params[6] = area
```

Part 3

```
# Call the C function
lib.solve_circle(params)

# Retrieve results
h, k, r, D, E, F = params[7], params[8], params[9], params[10],
    params[11], params[12]

# Print results
print("\n--- Circle Parameters ---")
print(f"Center (h, k): ({h:.3f}, {k:.3f})")
print(f"Radius: {r:.3f}")
print(f"Equation: x^2 + y^2 {D:+.3f}x {E:+.3f}y {F:+.3f} = 0")
```

Part 4

```
# Plot circle and diameter lines
theta = np.linspace(0, 2 * np.pi, 400)
x_circle = h + r * np.cos(theta)
y_circle = k + r * np.sin(theta)

# Lines
x_vals = np.linspace(-10, 10, 300)
y1 = (a1 * x_vals + c1) / (-b1)
y2 = (a2 * x_vals + c2) / (-b2)

plt.figure(figsize=(6, 6))
plt.plot(x_circle, y_circle, 'b', label='Circle')
plt.plot(x_vals, y1, 'r--', label=f'{a1}x {b1:+}y {c1:+} = 0')
plt.plot(x_vals, y2, 'g--', label=f'{a2}x {b2:+}y {c2:+} = 0')
```

Python Code: Plotting

Part 5

```
plt.scatter(h, k, color='black', s=50, label='Center (1,-1)')
plt.text(h, k, ' (1, -1)', fontsize=10, color='black', va='center',
        ha='left')
plt.title("Circle with Given Diameter Lines")
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.grid(True)
plt.legend()
plt.axis('equal')

# Save figure as fig13.png
plt.savefig("fig13.png", dpi=300)

# Show the figure
plt.show()
```


Graphical Representation

